Weigh Out – Game Design Document

Version 0.1 | October 2026 | Author: Parham Motaharimajd (Game Designer)

Status: playable demo in development, Steam release targeted for around March 2027. This is a living document: values are tuned through playtesting, and items marked **TBD** are not decided or not yet measured.

1. Overview

One-sentence pitch: A first-person multiplayer platformer where teams of wizards, imprisoned in an ancient prison, race to the top using weight-shifting magic, because only one clan can escape.

Item	Detail
Genre	First-person multiplayer platformer with action elements
Engine	Unreal Engine 5
Platform	PC (Steam)
Input	Keyboard and mouse; PC and Xbox controllers
Players	2–4 per match (1v1, 2v2, 3v1, 2v1); architecture allows up to 6
Tone	Cartoony, mysterious
Target audience	Competitive players who want to play against friends in a different kind of world
Price (planned)	About $7
Release (planned)	Around March 2027

Comparable games: a mix of Fall Guys-style and Crash-style racing with simple shooter gameplay.

Why wizards: wizards let the game support many different kinds of magic abilities without needing to justify weapons or gadgets. Players carry nothing; every tool is a spell.

2. Design Pillars

1. **Chaos:** matches should feel frantic and unpredictable.
2. **Mastery:** players should feel satisfied when they use their skills better than their opponents.
3. **Comeback:** a player or team that falls behind always has a way back.
4. **Tension:** rising lava, the contested final platform and weight management keep stress high.

Every feature should support at least one of these pillars. Features that don't are cut.

3. Core Gameplay

Goal: be the first team to escape the prison by reaching and holding the top platform. Win three rounds to win the match.

Core loop:

1. Escape the starting cell.
2. Climb the prison by managing your weight, using launchers and platforming.
3. Shoot, pull, shield and disrupt opponents to slow them down.
4. Reach the final platform and hold it for 15 seconds.
5. Win the round, then repeat until a team has 3 points.

Key idea: combat is part of movement, not separate from it. Shooting a rival makes you lighter and them heavier, so every fight directly changes how well each player can climb.

4. The Weight System

Weight is the central mechanic and acts as both the player's "health" and their mobility stat. Nobody is damaged or eliminated by shots.

Effects of low weight:

- Higher jumps
- Slower falls
- Launches have less effect on you, which protects you from being knocked off a platform

Effects of high weight:

- Shorter jumps, faster falls, easier to launch away

Ways to lose weight:

- Hitting another player with a shot (the shooter loses weight, the target gains it)
- Standing on a charger station

Ways to gain weight:

- Being hit by a shot
- Passing through lasers

Weight values: weight ranges from 0 (lightest) to 100 (heaviest). A basic shot makes the shooter lose 20 weight and the target gain 18. A homing projectile makes the target gain 60 and the shooter lose 30. Exact jump-height and fall-speed curves are still **TBD**. Weight is shown as a numeric value on a scale icon in the HUD.

Design intent: every shot is a choice. Hitting a rival helps you and hurts them, but it also costs mana and exposes you to return fire. Staying light is powerful, but being too light makes it hard to come back down when you want to.

4.1 Charger Stations

- Standing on a charger station gradually reduces weight.
- Every 2 seconds a quick-time event (QTE) appears. Pressing the button at the right moment gives an extra sudden reduction.
- Gives players a non-combat way to lose weight, so the climb is not only about shooting.

4.2 Lasers

- **Static lasers:** add 30 weight (30% of the full scale) once each time a player passes through them.
- **Pulsing lasers:** switch on and off and add weight continuously while a player stays inside the beam.
- Some lasers rotate or move along platforming sections.

5. Abilities

All abilities are magic. Players do not carry items. Abilities use one of two resource systems:

- **Cooldown abilities:** Shield, Pull, Dash

- **Mana abilities:** Basic shot, Launcher, Homing projectile

Mana regenerates when not in use (regeneration rate **TBD**). Costs: basic shot 25%, launcher about 50% (a full bar allows two back to back), homing projectile about 60%.

5.1 Basic Shot

- A magic projectile with no set range.
- Costs 25% of the mana bar.
- Does no damage: the shooter loses 20 weight and the target gains 18.

5.2 Launcher Bomb

- Uses mana.
- Explodes only on impact with a surface or player.
- Launches everyone inside the blast radius, away from the blast center. Launch power drops with distance from the explosion.
- Used to launch yourself up platforms, or to knock rivals off.
- Launch effect is reduced on lighter players.

5.3 Homing Projectile

- Must be picked up: a **white mist** marker in each level drops a homing projectile when a player passes through it.
- One use only; the player must return to a mist to get another.
- Costs about 60% of the mana bar.
- Easier to aim: it follows nearby players and bounces once off a wall, returning along its trail.
- On hit, the target gains 60 weight and the shooter loses 30.
- Role: comeback tool for trailing players.

5.4 Pull

- Uses a cooldown (3–7 seconds, depending on the ability).
- Three uses: (1) pull a teammate toward you when you are ahead so they can keep up, (2) pull rivals off the final platform or disrupt them, (3) pull yourself toward fixed pullable objects placed in the map, which speeds up movement and helps very light players get down.

- Range: about 25 meters. Pull force: **TBD**.

5.5 Shield

- Hold a button and move the mouse to draw; a VFX follows your view while drawing.
- On release, a thick cartoony pipe-like barrier appears along the drawn shape. It has collision.
- Thickness is about 50 cm (enough to cover a head and neck). Length depends on the drawing, from a small sphere up to almost a full circle. It always forms about 2 meters from the player who draws it.
- Blocks projectiles, shots and pulls, and can be drawn behind yourself so a launch pushes you into the shield instead of off the platform.
- Stays where it was drawn and disappears after a few seconds; only then can a new one be drawn. Cooldown: 3–7 seconds, depending on tuning.

5.6 Dash

- Uses a cooldown (3–7 seconds, depending on the ability).
- A simple burst that pushes the player about 5 meters in their moving direction.
- Mainly helps with platforming and movement speed.

6. Match Structure

- 2–4 players per match: 1v1, 2v2, 3v1 or 2v1. Up to 6 players is supported by the design.
- 3v1 and 2v1 use the same rules as the standard modes, with no special balancing.
- First team to **3 points** wins the match.
- A round ends when a team holds the final platform for 15 seconds.

Final platform rules:

- Holding the platform for 15 seconds wins the round.
- If an opposing player reaches the platform during the countdown, the timer stops and the teams fight; players try to launch each other off until only one team remains on the platform.

Lava:

- Once a player passes through the **red mist** on the upper part of the second floor, lava starts rising from the bottom of the first floor and everyone is notified.

- It takes roughly 2 minutes to reach the top of the third floor. At that point the final platform is the only safe place.
- A player who touches lava dies and spectates until the end of the round.
- If every player dies in lava, the round restarts.
- If only one team remains alive, that team must finish the round while the other team spectates. They can still fall in lava and score nothing.
- Purpose: falling used to just mean climbing again; the lava makes mistakes costly.

7. Levels

Players choose a level type in the lobby: **Vertical, Horizontal or Arena**. The game picks one of three random levels of that type (currently one level per type is built).

7.1 Vertical Level (flagship)

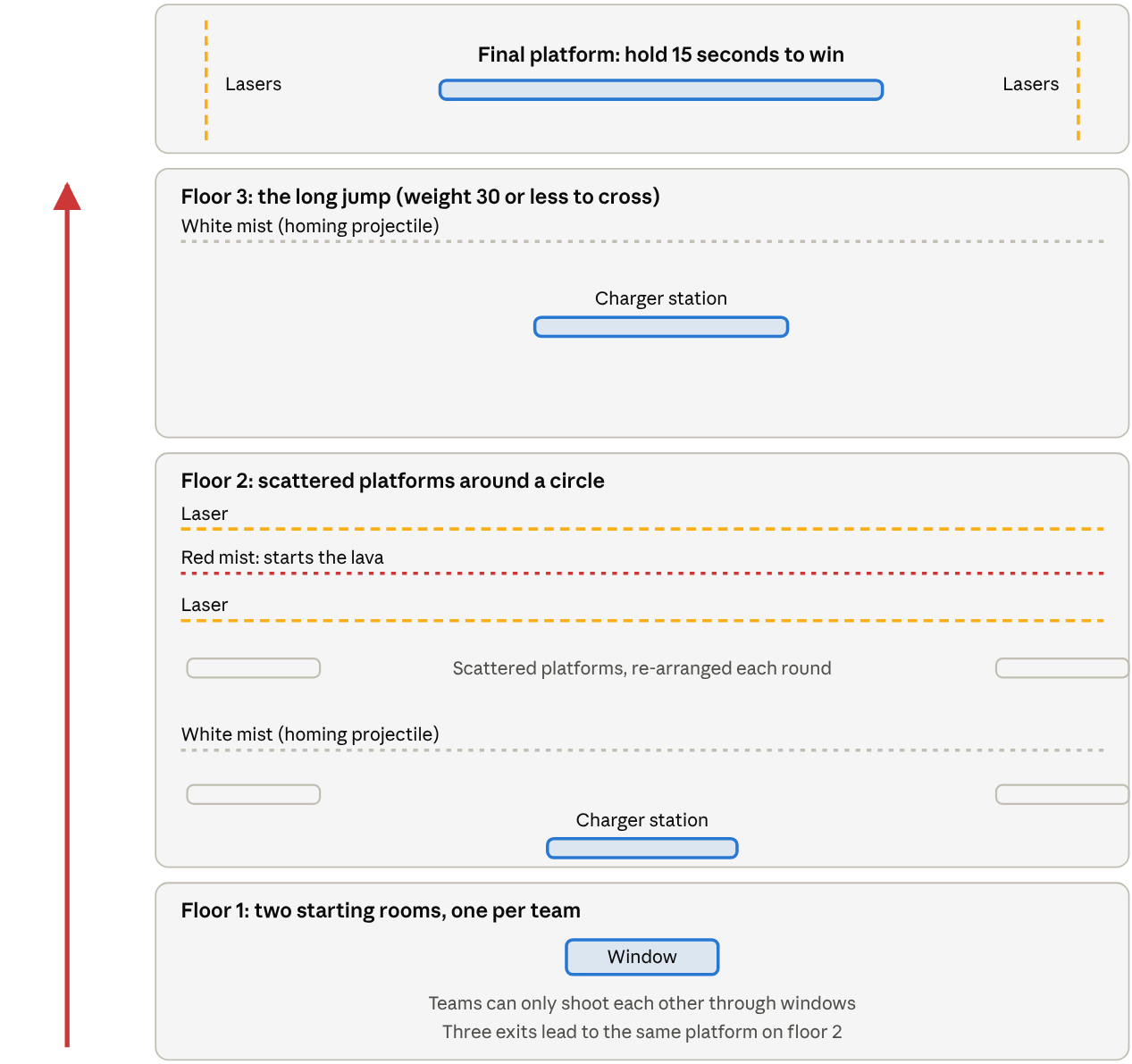
Floor 1: Starting rooms. Two half-circle rooms, one per team. Players start in sealed cells and must use a launcher on a breakable floor to escape, which also teaches the core mechanic. The only interaction between teams is shooting through windows in the center of each room. Three exits lead to the same platform.

Floor 2: First meeting and platforming. The platform where teams first meet. Small barrels provide cover for dodging shots. Above it, platforms are scattered around a circle, leaving the center empty so players can fight in the open. About 5 meters up is a charger station with a hole in it. Rotating and moving lasers sit along the route. A white mist (homing projectile) is placed low on the floor, and the red mist (lava trigger) is placed high.

Floor 3: The long jump. A flat circular floor with a charger station, lasers and a white mist. To reach the last platform, players must be at weight 30 or less (on the 0–100 scale) so a launcher jump can carry them across.

Final platform: a single platform in open space. The only obstacle is opposing players.

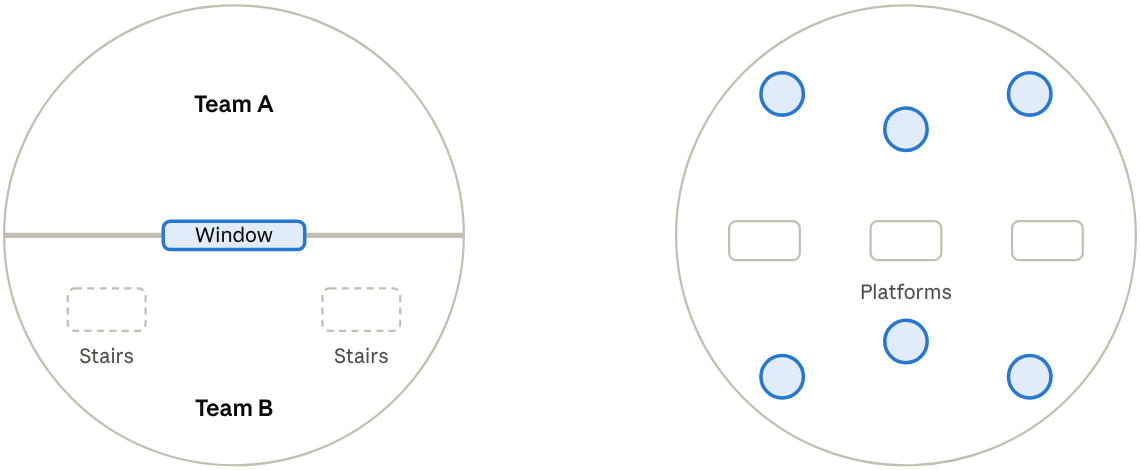
Randomization: the second-floor platforms are re-arranged each round so players can't memorize their jumps and launches. Most other geometry is hand-built.



vertical level side view · 3 floors and the final platform

Side view of the vertical level, drawn from the designer's sketch. Items appear in the order players meet them while climbing; distances are not to scale.

Teams start apart on floor 1 and first meet on floor 2



Floor 1: two half-circle rooms **Floor 2: open center for fights**

Circles on floor 2 mark ways down to the floor below

floors 1 and 2 top-down · schematic

Top-down schematics of floors 1 and 2, also from the sketch. Exact room and platform dimensions are not shown.

7.2 Horizontal Level

Same complexity and layers as the vertical level, but travel is sideways. It is designed as the easier level type because fighting and platforming are easier when you don't need fast mouse movement to go straight up.

7.3 Arena Level

One floor with holes and platforms. Players score by pushing opponents off the platform.

7.4 Demo Scope

The demo includes one vertical level. Horizontal and Arena are shown in the lobby as locked options to tease the full game.

8. UX, HUD and Onboarding

HUD:

- Weight bar: a large readable section plus one exact number, shown as a scale icon

- Crosshair
- Top bar showing which team is winning and each team's points
- Ability icons (Pull, Shield, Dash, Launcher) with the correct button for each
- Mana bar, with a homing projectile icon appearing on it when one is held

Onboarding:

- A narrator explains what the player can do; the rest is communicated through the HUD so players learn by doing.
- An opening camera sequence shows the goal at the start of each round.
- A launcher bomb icon on the HUD keeps the core movement tool in mind.
- The sealed-cell wall that breaks only with a launcher teaches the launch mechanic at the start of every match.

8.1 Controls (keyboard and mouse)

- **Move:** W, A, S, D
- **Jump:** Space
- **Shoot:** left mouse button
- **Draw shield:** right mouse button (hold and move the mouse)
- **Dash:** Shift
- **Pull:** C
- **Launcher:** E
- **Homing projectile:** Q
- **QTE events (charger station):** Control

Controls for PC and Xbox controllers are also supported; the controller layout is **TBD**.

9. Art, Audio and Tone

- **Visual style:** cartoony wizards and an ancient prison setting.
- **Music:** mysterious to match the theme, but fast-paced so players feel they are in a race.
- **SFX and VFX:** should match the feeling of the magical items and spells.

10. Playtesting Log

Round 1 (up to 4 testers: 2 casual players, 1 non-gamer, 1 professional FPS player)

- Finding: the loop was repetitive (shoot, get lighter, launch, climb).
- Changes: redesigned the level into escalating floors with distinct purposes; added charger stations; added lava; changed the win condition from touching the final platform to holding it for 15 seconds.

Round 2 (same tester profile)

- Findings: players did not know where to go, and forgot launcher bombs were their main movement tool.
- Changes: opening camera sweep; launcher HUD icon; launcher-only breakable cell wall; weight bar redesigned into a scale icon with numbers. Several bugs were also found and fixed.

Quantitative results (session lengths, round times, win rates): **TBD** after the next round.

11. Production

Team:

- Parham Motaharimajd: game design, level design, character, animation, VFX, SFX, UI and part of the Blueprint programming
- Armin Sadeghi: programmer (Blueprints)
- Mohammadjavad Momemni: environment art and assets
- Saman Rostami: social media (paid contractor, not a core team member)

Demo (current): vertical level, core mechanics and tools, tutorial, locked-level teasers.

Full release (planned): all three level types with three levels each.

Post-release (planned): solo mode, skin customization.

Release plan: Steam, around March 2027, about $7.

12. Open Questions

- Exact jump-height and fall-speed values at different weights
- Pull force, mana regeneration rate and the exact cooldown for each ability (all fall between 3 and 7 seconds)

- Controller button layout
- Quantitative playtest data from the next round (session length, round time, win rates)